

DEBAJIT GOSWAMI

FOUNDER & CHIEF EXECUTIVE OFFICER | LEAD AI ARCHITECT

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Technical Founder and AI Systems Architect focused on bridging the gap between elite engineering and enterprise-grade deployment. Specializing in multimodal infrastructure and autonomous agent orchestration, I lead teams to engineer high-concurrency backends and low-latency inference pipelines utilizing Groq LPUs and advanced LLM frameworks.

EXECUTIVE & TECHNICAL ARCHITECTURE

- System Architecture:** Multimodal Architectures, High-Concurrency Backends, Distributed Task-Queues (Celery/Redis).
- AI Infrastructure:** LLM Orchestration, Agentic Frameworks, Groq LPU Inference (t < 100ms), Vector DBs (pgvector).
- Core Stack:** Python (Expert), Go, C++, React/Node.js, PostgreSQL, Docker, AWS, Kubernetes.

FOUNDING & LEADERSHIP EXPERIENCE

Singularity Horizon Technologies Pvt. Ltd.

April 2025 - Present

Founder & Chief Executive Officer

- Strategic Vision:** Founded and incorporated the company to deliver enterprise-grade, highly concurrent AI infrastructure solutions to the B2B market.
- Product Architecture:** Architected 'Orion Helix AI', a proprietary multimodal infrastructure core enabling seamless real-time processing of disparate data streams.
- Team Leadership:** Manage a core engineering team of 4, directing the product roadmap from conceptual R&D to deployment, ensuring rigorous standards in systems design.
- Technical Execution:** Engineered autonomous agent frameworks utilizing Llama 3.3 and optimized Go/gRPC backends, reducing internal service latency by 40%.

National Hackathon (LPU) — Team Zero-Day Coders

March 2026

Lead Systems Developer | Project Lead

- Led the technical development of an Intelligent Code Review Bot featuring custom AST parsing for deep structural analysis of multi-language repositories.
- Directed full-stack implementation, coordinating Go for background tasks and React for the security dashboard to detect 0-day exploits in real-time.

KEY TECHNICAL INITIATIVES

Distributed Inference Pipeline

2025

- Engineered a scalable architecture using Celery and Redis to handle heterogeneous ML workloads across distributed clusters, enabling dynamic resource allocation for GPU/LPU nodes.

Orion Portfolio Engine (Enterprise Showcase)

2026

- Built a high-performance 3D interface using React-Three-Fiber integrated with a custom Express backend to demonstrate real-time AI chat inference capabilities to prospective clients.

EDUCATION

Lovely Professional University (LPU)

2025 - 2029 (Expected)

B.Tech in Computer Science and Engineering (Specialization in AI & ML)

- Active researcher in Agentic AI, balancing rigorous academic coursework with the demands of scaling a technology startup.